

TOP-K IS NOT A BUDGET FOR HYBRID RETRIEVAL

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ABSTRACT

Diverse queries and evolving corpora make fixed Top- L cutoffs difficult to transfer in hybrid retrieval for retrieval-augmented generation (RAG). Exact fusion avoids a fixed cutoff, but the reads needed to complete a fixed Top- K remain query-dependent. We present DiBud (Direct Budgeting), which takes a read budget and returns the exact prefix certified within it. DiBud maintains bounds on unread contributions and prioritizes the channel blocking the next output position. The budget directly controls reads; the available evidence determines the output length. Replays on five query sets show a 1.06%–5.95% improvement in normalized certified-prefix area over balanced reading. Across five corpus snapshots, output lengths vary as the corpus changes, and every delivered prefix matches the corresponding complete reciprocal rank fusion ranking.

Index Terms— Hybrid retrieval, reciprocal rank fusion, budgeted retrieval, exact prefix, RAG

1. INTRODUCTION

Hybrid retrieval for retrieval-augmented generation (RAG) commonly combines the Top- L results of dense and sparse retrievers using reciprocal rank fusion (RRF) [1, 2]. A fixed window limits reads, but also removes contributions beyond that window. When a document appears in one channel but not in the other channel’s window, fusion assigns zero to the latter contribution. Yet absence from the window only means that the rank has not been observed: its contribution is unknown. Replacing it with zero can change the output order even when the observed candidates already contain every document in the complete-fusion Top- K .

Diverse, open-ended requests and continuing corpus updates change channel rankings and their overlap, making a fixed L difficult to transfer across queries and corpus snapshots. Exact fusion avoids a fixed window by bounding unread contributions and reading until the first K documents and their order are determined [3]. EAHR implements this approach for hybrid retrieval [4]. However, the reads needed to complete a fixed Top- K can still vary sharply across queries.

Resolving a few contested positions can require reading far down the channel rankings.

Resource-constrained requests therefore need a direct limit on reads. Specifying K determines how many results must be completed, but does not bound the work required to complete them. We instead take a read budget as input and return the entire certified prefix at that budget. Execution determines the output length, while its order remains identical to complete fusion.

We study this budgeted exact-prefix formulation through DiBud (Direct Budgeting). DiBud bounds unread contributions, prioritizes the channel blocking the next position, and emits results as they become certain. It combines selective reading from SNRA [5] with incremental exact output from LARA-IN [6]. Replays on five query sets compare the certified yield of different schedules under equal budgets; a further replay across five corpus snapshots examines delivery as the corpus changes.

2. DIBUD: BUDGETED EXACT-PREFIX FUSION

Given a query, a corpus snapshot, and a read budget B , DiBud reads the dense and sparse rankings on demand and returns a certified prefix of their complete RRF ranking. The budget determines how much can be read; certification determines how much can be returned.

2.1. Budget and output

Let $r_i(x)$ denote the one-based rank of document x in channel $i \in \{D, S\}$, and let $g(r) = 1/(c+r)$ for rank constant $c \geq 0$. Set $r_i(x) = \infty$ and $g(\infty) = 0$ if x is outside that channel’s support. The complete fusion score is

$$F(x) = \sum_{i \in \{D, S\}} g(r_i(x)). \quad (1)$$

Sorting the union of the channel supports by decreasing $F(x)$, with a fixed document-identifier order to break ties, defines the complete sequence T^* .

Both channels supply resumable exact ranking prefixes from the same snapshot. If d_D and d_S are their read depths

and A is the output sequence, the required contract is

$$d_D + d_S \leq B, \quad A = T_{1:K}^*, \quad K = |A|. \quad (2)$$

K is not specified in advance. The output is empty if no position can be certified. The budget counts channel entries read, including separate reads of the same document in different channels.

2.2. Prefix certification

Let S_i be the documents already read from channel i . An unread contribution in that channel is bounded by

$$u_i = \begin{cases} 0, & \text{if confirmed exhausted,} \\ g(d_i + 1), & \text{otherwise.} \end{cases} \quad (3)$$

For an observed document x , the score bounds are

$$\ell(x) = \sum_{i:x \in S_i} g(r_i(x)), \quad (4)$$

$$h(x) = \ell(x) + \sum_{i:x \notin S_i} u_i. \quad (5)$$

A document unseen in both channels has score at most $h_\emptyset = u_D + u_S$.

Among observed documents not yet emitted, select x with the highest lower bound, breaking ties by the fixed identifier order. Certify x as the next output when its lower bound exceeds every remaining competitor's upper bound and $h_\emptyset$. Equality with an observed competitor is allowed only when x wins the identifier tie; comparison with the unseen bound remains strict. Append x and repeat until the next position cannot be certified.

The bounds ensure $\ell(x) \leq F(x) \leq h(x)$. Each emitted document therefore precedes every remaining document, including any not yet observed. Applying this argument at each output position establishes the prefix equality in (2).

2.3. Reading and stopping

When the next position cannot be certified, let y be the observed competitor with the highest upper bound, excluding x and the emitted prefix and breaking ties by identifier. If $h(y) \geq h_\emptyset$ and y is missing a contribution from exactly one channel, advance that channel. This either reveals the contribution or tightens its upper bound.

Otherwise, advance the channel whose next batch offers the larger reduction in the unread bound. For a batch of b entries in a channel that remains open after the read, this reduction is

$$\Delta_i = g(d_i + 1) - g(d_i + b + 1). \quad (6)$$

If exhaustion is established, the new bound is zero. Exhausted channels are skipped, and equal reductions are resolved by

a fixed channel order (dense first). This rule also initializes reading when no document has been observed.

After each batch, update the bounds and extend the certified prefix. The batch size is capped by the remaining budget. When the budget is consumed, or both channels are exhausted, return the accumulated output. Certification depends on the observed bounds, so changing the read schedule affects the number of certified results without changing the meaning of an emitted position.

3. EXPERIMENTS

3.1. Experimental setup

We use TREC-DL 2019/2020 [7, 8], NFCorpus [9], SciFact [10], and TREC-COVID [11], with 43, 54, 323, 300, and 50 queries, respectively (770 in total). The two TREC-DL sets share the MS MARCO passage corpus. We also replay the same 30 queries across five TREC-COVID corpus snapshots. Dense retrieval uses bge-small-en-v1.5 [12]; sparse retrieval uses BM25 [13] with $k_1 = 0.9$ and $b = 0.4$. Equal-weight RRF uses the frozen implementation's rank contribution $1/(59 + r)$.

All methods replay the same exact channel prefixes, with at most 64 entries per batch and total read budgets

$$\mathcal{B} = \{128, 256, 512, 1024, 2048, 4096, 8192, 10000\}.$$

We compare balanced reading, competition-pressure scheduling, and DiBud. They share the certification rule and differ only in channel selection. Balanced reading alternates channels. Competition pressure weights each channel's bound reduction by the number of unresolved competitors missing that channel's contribution; the unseen bound also contributes when it blocks certification. Pressure ties favor the shallower channel, then the dense channel.

The metric is certified output length $K(B)$, reported with caps of 20 and 100. These caps are reporting limits, not requested completion targets. To summarize the budget range, we integrate $\min(K(B), 100)/100$ against $\log B$ using the trapezoidal rule and divide by the log-budget span. We compute this normalized area for each query and then average within each set. Cross-set summaries weight the five sets equally.

All certified outputs are checked position by position against the complete RRF reference for their snapshot. Replay and reference use the same floating-point scoring and identifier tie order. Reaching the end of a saved prefix does not establish channel exhaustion; an unconfirmed channel retains its unread bound. The experiments measure logical ranking reads.

3.2. Certified yield at equal budgets

Table 1 shows that DiBud improves normalized certified-prefix area over balanced reading on all five query sets, by

Table 1. Normalized certified-prefix area over the read-budget range, capped at 100 results. Higher is better.

Query set	Balanced	Pressure	DiBud
TREC-DL 2019	0.3456	0.3629	0.3629
TREC-DL 2020	0.2976	0.3112	0.3112
NFCorpus	0.5714	0.5774	0.5775
SciFact	0.4170	0.4367	0.4368
TREC-COVID	0.2204	0.2334	0.2335

Table 2. Mean certified count for the same 30 queries across five corpus snapshots, with a budget of 2048 reads and a reporting cap of 20.

Snapshot	Balanced	DiBud
Round 1	17.70	17.93
Round 2	17.20	17.53
Round 3	16.03	16.37
Round 4	15.30	15.63
Round 5	14.83	15.10

1.06%–5.95% relative. Its results are close to competition-pressure scheduling.

At a budget of 2048 reads, the mean certified count among the first 100 positions increases from 41.77 with balanced reading to 44.87 with DiBud, a gain of approximately 7.4%. Among the first 20 positions, it increases from 18.07 to 18.17. Selective reading yields more certified results under the same budget; at this budget, most of the gain occurs in longer prefixes.

3.3. Delivery across corpus snapshots

With queries and budget held fixed, DiBud’s mean certified count decreases from 17.93 in the first snapshot to 15.10 in the fifth (Table 2). It exceeds balanced reading in all five snapshots.

The same read budget certifies different numbers of results as the corpus changes. DiBud meets that budget by returning prefixes of different lengths, each matching the complete fusion ranking of its corresponding snapshot.

4. CONCLUSION

Diverse queries and evolving corpora make fixed L difficult to transfer in hybrid retrieval for RAG, while completing an exact Top- K still requires a variable number of reads. DiBud takes the read budget directly, bounds unread contributions, and returns the prefix certified within that budget. The output length follows the available evidence. Replays on five query sets show that selective reading increases certified yield at equal budgets. Across corpus snapshots, output lengths

change while every delivered prefix remains consistent with the corresponding complete RRF ranking.

5. ACKNOWLEDGMENTS

OpenAI Codex assisted with drafting, English translation, and language editing of the abstract and Sections 1–4 from the research notes and Chinese manuscript.

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